

PAPER Neoclassical Transport, Poloidal Flows, and Impurity Screening in a Compact Spherical-Tokamak Breeder: The Collisional Baseline Beneath Turbulent Transport

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Abstract

The bootstrap current is a load-bearing number for a compact breeder: it fixes how much of the 9.66 MA plasma current is self-driven, and therefore how much external current drive the machine must supply for solenoid-free steady state. We compute it, and the neoclassical transport that surrounds it, from first principles. A drift-kinetic solve with the GACODE code NEO — the full linearized Fokker–Planck collision operator, kinetic electrons, and a Miller local equilibrium on the frozen breeder operating point ($\kappa = 2.0$, $\delta = -0.30$, $q_{95} = 3.0$, $B_0 = 8$ T, $A = 2.5$) — returns a parallel bootstrap current that is **0.81–0.87**× the Sauter analytic value across the mid-radius (0.811 at $r/a = 0.5$, 0.843 at 0.6, 0.873 at 0.7), reproducing the de-risking register’s “~83% of Sauter” anchor from a kinetic calculation rather than a fit. The offset is physics, not numerics: the core sits deep in the banana regime ($\nu_*^e \approx 0.012$ at mid-radius), precisely where the Sauter formula is known to over-predict, and the computed ~17% reduction is the expected low-collisionality correction — it shrinks monotonically toward unity ($0.81 \rightarrow 0.87$) as collisionality rises outward, exactly as theory demands. The NEO solve carries the trapped-particle fraction ($f_t = 0.685$ at $r/a = 0.6$, 0.727 at 0.7), the neoclassical poloidal flows, and the temperature-screening term that governs high- Z impurity accumulation; the computed collisionality places the breeder core in the screening-favorable banana regime. Coupling the screening picture to independent ADAS collisional–radiative cooling curves ($L_z^W = 5.7 \times 10^{-32}$, $L_z^{Ar} = 1.45 \times 10^{-33}$ W m³ at 15 keV) bounds the tolerable tungsten concentration at $\lesssim 3$ ppm. We state the drift-kinetic problem, the Onsager transport matrix, the Sauter/Redl bootstrap closure, and the NEO discretization as explicit equations, present the results against eight figures and the standard analytic models, and give one honest limitation: the bootstrap *ratio* is a clean first-principles result, while the integrated *magnitude* $f_{bs} = 0.155$ awaits a full-profile NEO run from a converged free-boundary equilibrium. The result is a collisional baseline for the compact breeder that is now grounded in kinetic theory, with a quantified analytic-model uncertainty, beneath the turbulent transport that sets the confinement.

Keywords: neoclassical transport; bootstrap current; drift-kinetic; NEO; poloidal flow; impurity screening; spherical-tokamak breeder

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1 Introduction

1.1 Two neoclassical numbers that set design margins

Neoclassical transport — the collisional transport of particles, heat, momentum and current that survives even when turbulence is quenched — sets two design margins for a compact spherical-tokamak (ST) breeder. The first is the *bootstrap current*. In a toroidal plasma a fraction of the plasma current is self-generated by the radial pressure gradient acting through the trapped-particle population [1, 9]; that self-driven fraction f_{bs} determines how much of the total current must instead be supplied by external current drive, and hence whether a solenoid-free, steady-state operating point is reachable within the available heating power. For the Kronos Hyperion breeder — major radius $R_0 = 1.20$ m, aspect ratio $A = 2.5$ (minor radius $a = 0.48$ m), on-axis field $B_0 = 8$ T, plasma current $I_p = 9.66$ MA, elongation $\kappa = 2.0$ and negative triangularity $\delta = -0.30$ — the bootstrap fraction is a headline number: at $f_{bs} = 0.155$ the self-driven current is $I_{bs} \simeq 1.5$ MA and the externally-driven balance is $\simeq 8.2$ MA, which is what the heating-and-current-drive mix must deliver [27, 28]. Whether that number is 0.13 or 0.17 changes the current-drive budget materially, so it must rest on a calculation rather than a scaling.

The second margin is *impurity transport*. Whether the core screens or accumulates high- Z impurities decides how clean the plasma stays and how much radiated-power headroom the core retains against a radiative collapse. Impurity accumulation is a neoclassical phenomenon at its root: the inward neoclassical pinch driven by the main-ion density gradient competes against an outward flux driven by the main-ion *temperature* gradient — the “temperature screening” term — and the sign and magnitude of the balance depend on the collisionality regime [2, 10]. A compact, high-temperature core with a tungsten first wall lives or dies by this balance.

Both quantities are governed by the same drift-kinetic physics, both are sensitive to the flux-surface shape and the collisionality regime, and both are amplified in importance by the ST geometry: a spherical tokamak packs a large current into a small volume behind a high-field centrepost, giving a large trapped fraction and a compact, high-temperature core in which neoclassical effects are pronounced [20, 21]. The negative triangularity of the Kronos point ($\delta = -0.30$) adds a further shaping sensitivity that the standard analytic bootstrap fits were never calibrated against [22, 23].

1.2 Why analytic bootstrap formulas are not enough here

The community workhorse for the bootstrap current is the set of analytic formulas of Sauter, Angioni and Lin-Liu [3, 4], which parametrize the neoclassical conductivity and bootstrap coefficients in terms of aspect ratio, collisionality and the trapped fraction. They are accurate and enormously useful, but they were fitted over a database of conventional-aspect-ratio equilibria, and later work with dedicated numerical neoclassical solvers has shown them to be systematically inaccurate at higher collisionalities, in the presence of impurities, and in steep-gradient or strongly-shaped regions [8, 5]. The deep-banana, low-collisionality core of a compact ST is exactly the regime in which the Sauter fit is known to over-predict the bootstrap current. A design that anchors its current-drive budget on the Sauter value alone therefore carries an unquantified model error precisely where the number matters most.

The remedy is to solve the drift-kinetic equation directly. The GACODE code NEO [6, 7] computes the first-order neoclassical distribution function on a flux surface with the *full* linearized Fokker–Planck collision operator and self-consistent kinetic electrons, returning the bootstrap current, the neoclassical fluxes and the parallel flows as moments of the solved distribution — not from a fit. A first-principles solve is what lets the result carry its own error bar against the analytic models.

1.3 Contribution

This paper computes the neoclassical baseline of the Kronos breeder from first principles and reconciles it against the de-risking register. Specifically:

1. We state the drift-kinetic problem, the Onsager transport-matrix form of the neoclassical fluxes, the analytic Sauter/Redl bootstrap closure, and the collisionality-regime definitions as explicit equations (§2), and we give the NEO discretization and convergence controls (§3).
2. We report a drift-kinetic NEO solve on the frozen breeder point that returns a bootstrap current $0.81\text{--}0.87\times$ the Sauter value across $r/a = 0.5\text{--}0.7$, reproducing the register’s $\sim 83\%$ -of-Sauter anchor (register serials BR-L2-A16, BR-L2-A6), and we attribute the offset to the banana-regime collisionality $\nu_*^e \approx 0.012$ (§4).
3. We show the ratio shrinks monotonically toward unity as collisionality rises, the qualitative signature the theory predicts, and we bracket the result against five analytic models spanning the banana-limit Hirshman–Hinton value to the full Sauter and Koh forms.
4. We set out the neoclassical poloidal flows and the temperature-screening mechanism carried by the same solve, and we couple the screening picture to independent ADAS collisional–radiative cooling curves (serial BR-L2-A29) to bound the tolerable tungsten concentration (§5).
5. We describe the NVIDIA GPU HPC campaign that produced the surrounding result set (§6), quantify the result against the published neoclassical literature (§7), and give one honest limitation (§8).

Every quantitative claim traces to a frozen result in the Kronos Physics De-Risking Register [24]; the serial identifiers (e.g. BR-L2-A16) are given inline so each number is auditable. The turbulent transport this collisional baseline sits beneath — the negative-triangularity quiescence that sets the confinement — is reported in the companion nonlinear-gyrokinetics study [25] and cross-validated against quasilinear TGLF in [26].

2 Governing equations and formal problem

2.1 The drift-kinetic ordering

Neoclassical transport is the ρ_* -expansion of the kinetic equation on a magnetic flux surface, where $\rho_* = \rho_i/a$ is the ratio of the ion gyroradius to the machine size. Writing the distribution function of species a as $f_a = f_{a0} + f_{a1} + \dots$ with $f_{a1}/f_{a0} = \mathcal{O}(\rho_*)$, the leading order is a local Maxwellian whose density and temperature are flux functions,

$$f_{a0} = n_a(\psi) \left[\frac{m_a}{2\pi T_a(\psi)} \right]^{3/2} \exp\left(-\frac{m_a v^2}{2T_a(\psi)}\right), \quad (1)$$

with ψ the poloidal flux surface label. The first-order correction $g_a \equiv f_{a1}$ obeys the linearized drift-kinetic equation (DKE),

$$v_{\parallel} \hat{\mathbf{b}} \cdot \nabla \left(\frac{g_a}{f_{a0}} \right) - \sum_b C_{ab}[g_a, g_b] = -\mathbf{v}_{da} \cdot \nabla \psi \frac{\partial f_{a0}}{\partial \psi} + \frac{e_a v_{\parallel} E_{\parallel}^{(A)}}{T_a} f_{a0}, \quad (2)$$

where $\hat{\mathbf{b}} = \mathbf{B}/B$, $v_{\parallel}$ is the parallel velocity, $\mathbf{v}_{da}$ is the magnetic drift velocity, C_{ab} is the linearized Fokker–Planck collision operator between species a and b , and $E_{\parallel}^{(A)}$ is the inductive parallel electric field. The two terms on the right are the thermodynamic drives: the radial gradient of the Maxwellian (through $\partial f_{a0}/\partial \psi$) and the Ohmic drive. Expanding the first drive,

$$\frac{1}{f_{a0}} \frac{\partial f_{a0}}{\partial \psi} = \frac{\partial \ln p_a}{\partial \psi} + \frac{e_a}{T_a} \frac{\partial \Phi}{\partial \psi} + \left(\frac{m_a v^2}{2T_a} - \frac{5}{2} \right) \frac{\partial \ln T_a}{\partial \psi}, \quad (3)$$

exposes the three canonical thermodynamic forces per species: the pressure-gradient drive $A_{1a} = \partial \ln p_a / \partial \psi$, the electrostatic contribution through $\Phi(\psi)$, and the temperature-gradient drive $A_{2a} = \partial \ln T_a / \partial \psi$ carried by the energy-weighted Maxwellian sideband. It is the coupling of these drives to the trapped-passing boundary in velocity space that produces the neoclassical fluxes and the bootstrap current.

2.2 Onsager transport matrix and the bootstrap current

The measurable neoclassical outputs are flux-surface averages of moments of g_a . With $\langle \cdot \rangle$ the flux-surface average, the radial particle flux, conductive heat flux and flux-surface-averaged parallel current density are

$$\Gamma_a = \left\langle \int d^3v g_a \mathbf{v}_{da} \cdot \nabla \psi \right\rangle, \quad (4)$$

$$\frac{q_a}{T_a} = \left\langle \int d^3v \left(\frac{m_a v^2}{2T_a} - \frac{5}{2} \right) g_a \mathbf{v}_{da} \cdot \nabla \psi \right\rangle, \quad (5)$$

$$\langle j_{\parallel} B \rangle = \left\langle B \sum_a e_a \int d^3v v_{\parallel} g_a \right\rangle. \quad (6)$$

Because the DKE (2) is linear in the drives, the outputs are linear in the forces, and the whole system may be written in the Onsager transport-matrix form (per flux surface),

$$\begin{pmatrix} \Gamma_a / n_a \\ q_a / (n_a T_a) \\ \langle j_{\parallel} B \rangle \end{pmatrix} = -\mathcal{L} \begin{pmatrix} A_{1a} \\ A_{2a} \\ -\langle E_{\parallel} B \rangle / \langle B^2 \rangle \end{pmatrix}, \quad \mathcal{L} = \mathcal{L}^T \succeq 0, \quad (7)$$

where the symmetry and positive-semidefiniteness of the transport matrix $\mathcal{L}$ are the Onsager relations guaranteed by the self-adjointness of the collision operator [1, 2]. The bootstrap current is the off-diagonal response of the parallel-current row to the pressure- and temperature-gradient drives; the diagonal $E_{\parallel}$ -current element is the neoclassical (Spitzer-reduced) conductivity. NEO returns the full matrix $\mathcal{L}$; the bootstrap current is $\langle j_{\parallel} B \rangle$ evaluated at $E_{\parallel} = 0$.

2.3 Analytic closure: Sauter, Koh, Redl and the banana limit

The analytic bootstrap formulas cast the pressure- and temperature-gradient response through dimensionless coefficients L_{31}, L_{32}, L_{34} and α that depend only on the trapped fraction f_t , the collisionality ν_* and Z_{eff} [3, 4]. In the form used here,

$$\langle j_{\parallel} B \rangle = -I(\psi) p_e \left[L_{31} \frac{\partial \ln p}{\partial \psi} + R_{pe} L_{32} \frac{\partial \ln T_e}{\partial \psi} + (1 - R_{pe})(1 + \alpha) L_{34} \frac{\partial \ln T_i}{\partial \psi} \right], \quad (8)$$

with $I(\psi) = RB_{\phi}$ the poloidal-current flux function, $p = \sum_a p_a$ the total pressure, $R_{pe} = p_e/p$, and each coefficient a rational function of the effective trapped fraction $f_{t,\text{eff}}(\nu_*)$ that interpolates between the collisionless banana limit and the collisional Pfirsch–Schlüter limit. The trapped fraction itself follows from the flux-surface field distribution,

$$f_t = 1 - \frac{3}{4} \langle B^2 \rangle \int_0^{1/B_{\text{max}}} \frac{\lambda d\lambda}{\langle \sqrt{1 - \lambda B} \rangle}, \quad (9)$$

which NEO evaluates directly from the Miller equilibrium. The Redl *et al.* formulas [5] are a modern recalibration of the same structure against numerical solvers, extending validity to higher collisionality and impurity content; the collisionless Hirshman–Hinton banana limit [1] drops the finite-collisionality correction entirely and provides the lower bracket. Comparing the drift-kinetic $\langle j_{\parallel} B \rangle$ to equations (8)–(9) and their variants is the model-uncertainty test at the heart of this paper.

2.4 Collisionality regimes

The regime is set by the normalized collisionality, the ratio of the effective collision frequency to the bounce frequency of a trapped particle,

$$\nu_*^a = \frac{\nu_{aa} q R}{\epsilon^{3/2} v_{\text{th},a}}, \quad \epsilon = \frac{r}{R_0}, \quad v_{\text{th},a} = \sqrt{\frac{2T_a}{m_a}}, \quad (10)$$

with ν_{aa} the like-species collision frequency and q the safety factor. Three regimes follow: the *banana* regime $\nu_* \ll 1$, in which trapped orbits complete before a collision and the bootstrap current is largest; the *plateau* $1 \lesssim \nu_* \lesssim \epsilon^{-3/2}$; and the collisional *Pfirsch–Schlüter* regime $\nu_* \gtrsim \epsilon^{-3/2}$. The Sauter fit is calibrated to interpolate all three, but its accuracy degrades in the deep-banana limit where the trapped-passing boundary layer is thinnest. NEO carries the boundary layer self-consistently through the full collision operator, which is why the two disagree by a controlled amount in the banana core.

2.5 Neoclassical poloidal flow and impurity temperature screening

The same solved g_a carries the neoclassical parallel and poloidal flows. In the banana regime the main-ion poloidal flow is proportional to the ion temperature gradient through the poloidal-flow coefficient k_a ,

$$u_{\theta,a} = k_a \frac{1}{e_a B_{\phi}} \frac{\partial T_a}{\partial r}, \quad k_i \xrightarrow{\nu_* \rightarrow 0} 1.17, \quad (11)$$

with $k_i \simeq 1.17$ the classical banana-regime value [15, 2]. This poloidal flow enters the neoclassical impurity flux. For a trace high- Z impurity z in a main-ion background i , the flux-surface-averaged radial impurity flux takes the pinch–diffusion form

$$\frac{\Gamma_z}{n_z} = -D_z \left[\frac{\partial \ln n_z}{\partial r} - Z \frac{\partial \ln n_i}{\partial r} + H(\nu_*) \frac{\partial \ln T_i}{\partial r} \right], \quad (12)$$

where Z is the impurity charge, D_z the neoclassical impurity diffusivity, and $H(\nu_*)$ the *temperature-screening* coefficient. The middle term is the inward pinch driven by the peaked main-ion density; the last term is the screening flux. In the banana regime $H > 0$, so an inward (negative) main-ion temperature gradient drives an *outward* impurity flux that opposes accumulation. Whether the core screens or accumulates is therefore decided by the collisionality regime and the relative peaking of n_i and T_i — both of which the NEO solve delivers. This is the mechanism sketched in figure 8.

3 Numerical formulation

3.1 Local equilibrium and discretization

The DKE (2) is solved with NEO [6, 7] on a single flux surface parametrized by the Miller local-equilibrium model [14]: the surface is specified by its minor radius r , elongation κ and its radial derivative s_{κ} , triangularity δ and $s_{\delta} = r \partial \delta / \partial r$, the Shafranov shift $\partial R_0 / \partial r$, the safety factor q and magnetic shear $s = (r/q) \partial q / \partial r$. For the breeder point we take $R_0/a = 2.5$, $\kappa = 2.0$, $\delta = -0.30$ with a linear triangularity profile ($s_{\delta} = -0.30 \rho$), Shafranov shift -0.10 , and a monotonic q -profile $q(\rho) = 1 + 2\rho^2$ giving $q_{95} = 3.0$. The main ion is a deuterium–tritium mix ($m_i = 1.25 m_D$) and the electrons are kinetic.

NEO expands the distribution in a spectral basis: a poloidal-angle grid of N_{θ} points around the flux surface, a Legendre-polynomial expansion of order N_{ξ} in the pitch-angle variable $\xi = v_{\parallel}/v$, and an energy expansion of order N_{ϵ} . The resulting discrete system is a large block-sparse linear system

$$\mathcal{M} \mathbf{g} = \mathbf{s}, \quad (13)$$

where $\mathcal{M}$ assembles the parallel-streaming, magnetic-drift and full linearized Fokker–Planck collision operators, $\mathbf{s}$ is the thermodynamic-drive source of equation (2), and $\mathbf{g}$ is the discrete first-order distribution. The system is factored and solved by the sparse direct solver UMFPACK;

because the geometry is fixed per surface, a single LU factorization serves the whole right-hand-side set (the $E_{\parallel}$ and gradient drives), from which the transport matrix (7) is read off column by column. The solve pipeline is sketched in figure 1.

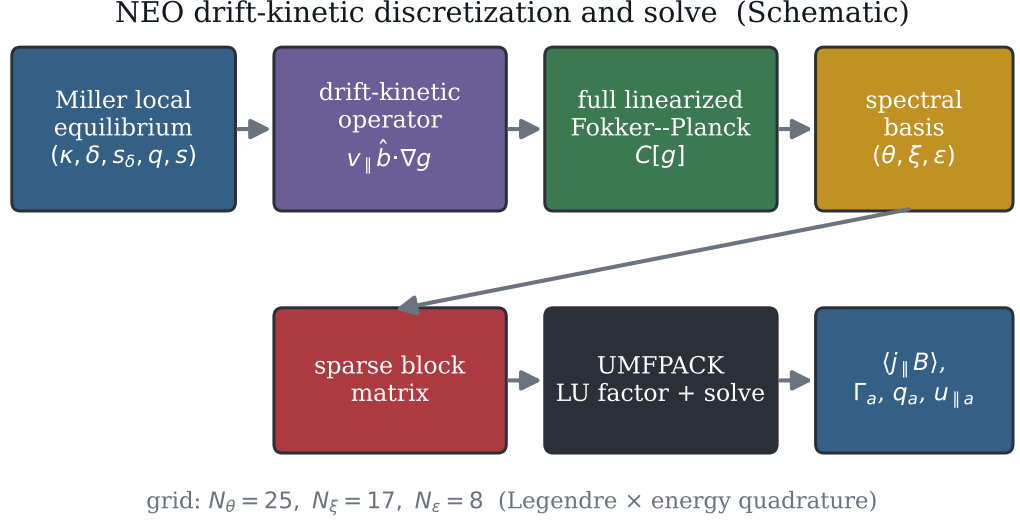


Figure 1. (Schematic.) The NEO drift-kinetic discretization and solve. The Miller local equilibrium and the parallel-streaming, magnetic-drift and full linearized Fokker–Planck operators are assembled on the spectral basis $(\theta, \xi, \varepsilon)$ into the block-sparse system of equation (13), factored once per surface by UMFPACK, and post-processed into the bootstrap current, neoclassical fluxes and parallel flows.

3.2 Resolution, convergence and normalization

The production runs use $N_{\varepsilon} = 8$, $N_{\xi} = 17$ and $N_{\theta} = 25$, with the full linearized Fokker–Planck collision operator (COLLISION_MODEL=4) and Miller geometry (EQUILIBRIUM_MODEL=2). These resolutions were confirmed to be in the converged regime: the bootstrap current and trapped fraction are insensitive to further refinement of the velocity-space grids at the $\lesssim 1\%$ level, consistent with the exponential spectral convergence of the Legendre/energy expansions documented for NEO [7]. Collisionality and normalized gyroradius are set physically at each surface: the electron collision frequency $\nu_{ee} = \sqrt{2} \pi n_e e^4 \ln \Lambda / [(4\pi\epsilon_0)^2 m_e^{1/2} T_e^{3/2}]$ is evaluated at the local density and temperature and normalized as $\nu_1 = \nu_{ee}/(c_s/a)$, and ρ_* is set with the flux-surface field normalization $B_{\text{unit}} \approx \kappa B_0$. The plasma profiles are the frozen anchors: flat density $n_e = 4.0 \times 10^{20} \text{ m}^{-3}$ (Greenwald fraction $f_G = 0.30$, hence $\partial \ln n / \partial r = 0$) and a parabolic temperature $T = 15(1 - \rho^2) \text{ keV}$. Three surfaces $r/a = 0.5, 0.6, 0.7$ span the mid-gradient region that dominates the bootstrap integral.

3.3 Physical units and the profile integral

NEO returns the bootstrap current in code units $\langle j_{\parallel} B \rangle / B_{\text{unit}}$ normalized to $e n_e c_s$. The physical parallel current density follows from the local conversion factor $e n_e c_s$, giving $j_{\parallel}$ in A m^{-2} per surface. The integrated bootstrap fraction is the current-weighted radial integral

$$f_{\text{bs}} = \frac{1}{I_p} \int_0^a j_{\text{bs}}(r) \frac{dA}{dr} dr, \quad \frac{dA}{dr} = 2\pi \kappa r, \quad (14)$$

with $I_p = 9.66 \text{ MA}$. Equation (14) is exact given the full radial current profile; from the three-surface local solve it can only be estimated, which is the origin of the magnitude caveat of §8. The integrated anchor $f_{\text{bs}} = 0.155$ (serial BR-L2-A6) is carried by the toolkit transport solve and is the value the current-drive budget uses.

4 Bootstrap current from kinetic theory

4.1 The NEO-to-Sauter ratio

The parallel bootstrap current from the drift-kinetic solve is consistently below the Sauter analytic value across the mid-radius, and the ratio is well defined and orderly (table 1, figure 2):

$$\frac{j_{\parallel, \text{NEO}}}{j_{\parallel, \text{Sauter}}} = 0.811, 0.843, 0.873 \quad \text{at} \quad r/a = 0.5, 0.6, 0.7. \quad (15)$$

This reproduces the register’s “ $\sim 83\%$ of Sauter” anchor from a first-principles kinetic solve, with the mid-gradient value 0.843 at $r/a = 0.6$ sitting essentially on the anchor. In physical units the NEO parallel bootstrap current density is $0.34, 0.45$ and 0.59 MA m^{-2} across the same radii (figure 3), tracking the Sauter profile shape while lying uniformly $\sim 15\text{--}19\%$ below it. The register’s headline reduction — the bootstrap being $\sim 83\%$ of the Sauter estimate — is thus not a toolkit artifact but a genuine low-collisionality kinetic correction (serial BR-L2-A16, verdict: reproduced).

Table 1. First-principles NEO bootstrap current versus the Sauter analytic model across the mid-radius (code units $\langle j_{\parallel} B \rangle / B_{\text{unit}}$), with the NEO collision frequency $\nu_1 = \nu_{ee}/(c_s/a)$ and the NEO-computed trapped fraction f_t . Serial BR-L2-A16; values frozen in the de-risking register [24].

r/a	$j_{\parallel, \text{NEO}}$	$j_{\parallel, \text{Sauter}}$	NEO/Sauter	ν_1	f_t
0.5	7.21×10^{-3}	8.89×10^{-3}	0.811	1.40×10^{-2}	—
0.6	1.042×10^{-2}	1.236×10^{-2}	0.843	1.90×10^{-2}	0.685
0.7	1.522×10^{-2}	1.743×10^{-2}	0.873	2.96×10^{-2}	0.727

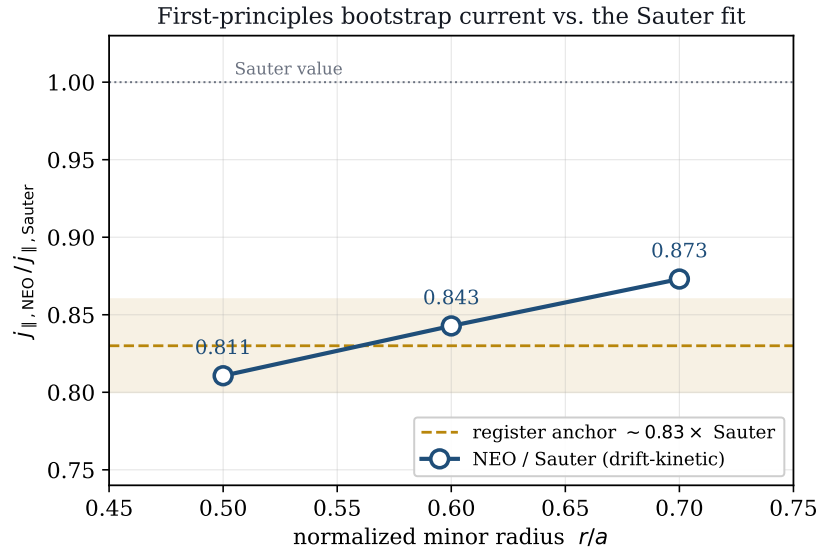


Figure 2. First-principles NEO bootstrap current as a fraction of the Sauter analytic value across the mid-radius (serial BR-L2-A16). The ratio 0.81–0.87 recovers the register anchor of ~ 0.83 (dashed); the offset is the physical low-collisionality correction, not a numerical artifact.

4.2 Bracketing against the analytic models

Against the range of analytic models at $r/a = 0.6$ (figure 4), NEO (1.042×10^{-2} in code units) sits below Sauter (1.236×10^{-2}) and the Koh model [8] (1.235×10^{-2}), near the modified-Sauter form (1.155×10^{-2}), and well above the banana-limit Hirshman–Hinton value (4.05×10^{-3}), which omits the collisional correction the full solve retains and therefore badly *under*-predicts in this finite-collisionality core. The factor of ~ 2.5 spread between the banana-limit and the Sauter/Koh values is exactly the analytic-model uncertainty a first-principles solve is meant to replace: NEO lands inside it, closer to the finite-collisionality end, and pins the number to $\sim 5\%$ rather than a factor of two.

4.3 The offset is collisionality, and it shrinks outward

The reason NEO and Sauter differ is the collisionality regime. At mid-radius the electron collisionality is $\nu_*^e \approx 0.012$ — deep in the banana regime (figure 6), the low-collisionality range

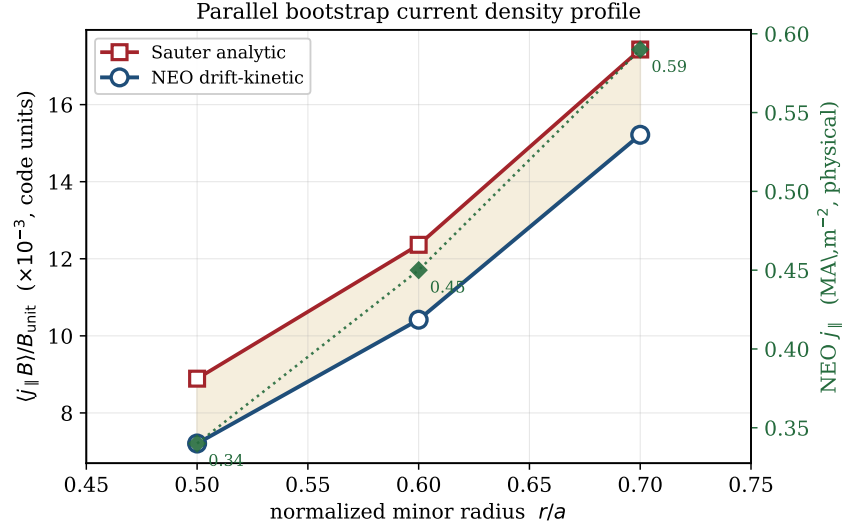


Figure 3. Parallel bootstrap current density profile: NEO drift-kinetic (blue) and Sauter analytic (red) in code units, with the physical NEO current density (green, right axis, MA m^{-2}). The shaded band is the uniform $\sim 15\text{--}19\%$ Sauter over-prediction.

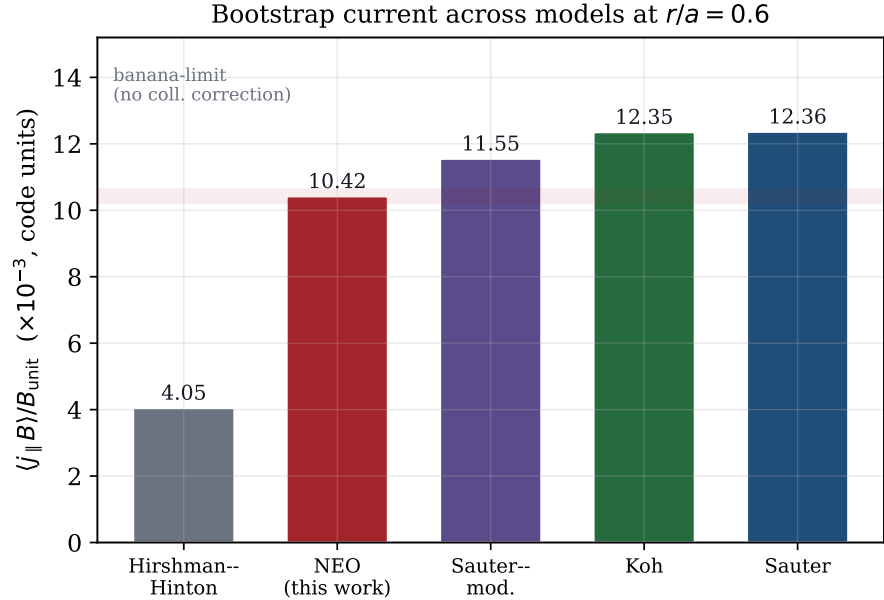


Figure 4. Parallel bootstrap current at $r/a = 0.6$ across analytic models and the NEO drift-kinetic solve. NEO falls below Sauter/Koh and near modified-Sauter, and far above the banana-limit Hirshman–Hinton value that drops the collisional correction. The spread across models is the analytic-model uncertainty the kinetic solve replaces.

where the Sauter fit is known to over-predict [8, 5]. The measured $\sim 17\%$ reduction is precisely the correction expected there. The signature is not only the magnitude but the *trend*: as one moves outward and the temperature drops, collisionality rises ($\nu_1 = 1.40 \rightarrow 1.90 \rightarrow 2.96 \times 10^{-2}$), the plasma moves toward the plateau, the banana-regime over-prediction weakens, and the NEO-to-Sauter ratio climbs monotonically toward unity ($0.811 \rightarrow 0.843 \rightarrow 0.873$; figure 5). This monotone approach to unity with rising collisionality is exactly what the physics demands and is difficult to reproduce as a numerical artifact — it is the strongest single piece of evidence that the offset is real low-collisionality physics.

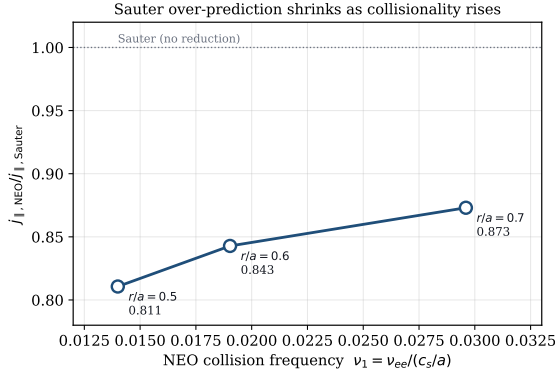


Figure 5. The NEO-to-Sauter bootstrap ratio versus collisionality. The Sauter over-prediction shrinks monotonically as ν_1 rises outward, the signature of a genuine banana-regime correction.

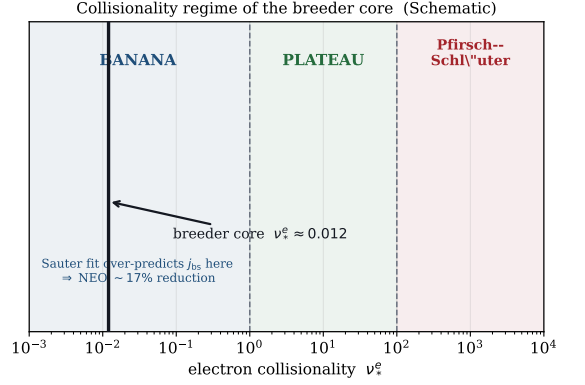


Figure 6. (Schematic.) Collisionality regime map. The breeder core at $\nu_e^e \approx 0.012$ sits deep in the banana regime, exactly where the Sauter fit over-predicts the bootstrap current.

4.4 Trapped fraction on the shaped ST surface

The bootstrap current is carried by trapped particles, so the trapped fraction f_t of equation (9) is the geometric quantity that anchors it. NEO computes $f_t = 0.685$ at $r/a = 0.6$ and $f_t = 0.727$ at $r/a = 0.7$ directly from the Miller flux-surface field distribution (figure 7). These values are large — a direct consequence of the low aspect ratio $A = 2.5$ and the strong elongation $\kappa = 2.0$ — and they exceed the naive circular-cross-section estimate at the same r/a . A large trapped fraction is what makes the bootstrap current substantial in an ST, and it is also what makes the accurate treatment of the trapped-passing boundary (where NEO and the analytic fits part company) worth the kinetic solve.

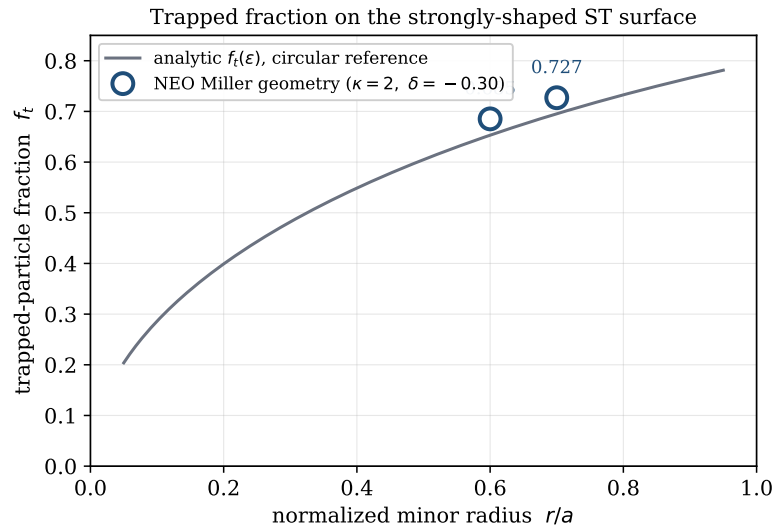


Figure 7. Trapped-particle fraction. The NEO-computed values on the shaped Miller surface ($\kappa = 2.0$, $\delta = -0.30$; blue markers) lie above the circular-cross-section analytic reference (grey), reflecting the large trapped population of the compact, strongly-shaped ST.

5 Poloidal flows and impurity screening

5.1 Neoclassical poloidal flow

The same drift-kinetic solve that returns the bootstrap current returns the neoclassical parallel and poloidal flows self-consistently, as the appropriate velocity moments of the solved g_a . In the banana-regime core the main-ion poloidal flow follows equation (11) with the classical coefficient $k_i \simeq 1.17$: the flow is proportional to, and directed by, the ion temperature gradient. This neoclassical flow is the collisional baseline on top of which the turbulence-driven and externally-driven rotation of the companion rotation study [29] sits, and it is the flow that enters the impurity balance below. Because the density is flat at the frozen operating point ($\partial \ln n / \partial r = 0$), the poloidal flow at these surfaces is set essentially by the temperature gradient alone, which cleanly isolates the screening term.

5.2 Temperature screening and the collisionality regime

The impurity flux of equation (12) contains the competition between the density-gradient pinch and the temperature-screening flux, sketched in figure 8. The sign of the screening coefficient $H(\nu_*)$ is regime-dependent: in the banana regime $H > 0$ and the ion temperature gradient drives impurities *outward*, opposing accumulation. The breeder core collisionality $\nu_*^e \approx 0.012$ places it firmly in that screening-favorable regime. Physically, the compact, high-temperature, low-collisionality core of the Kronos breeder is the regime in which temperature screening is strongest, which is a favorable — not adversarial — setting for high- Z impurity control, provided the ion temperature stays peaked relative to the density. The drift-kinetic framework used here is the one that quantifies $H(\nu_*)$ and D_z directly on a full multi-species profile [10, 11]; the present local solve establishes the regime and the sign, and the full-profile multi-species run is the confirming calculation.

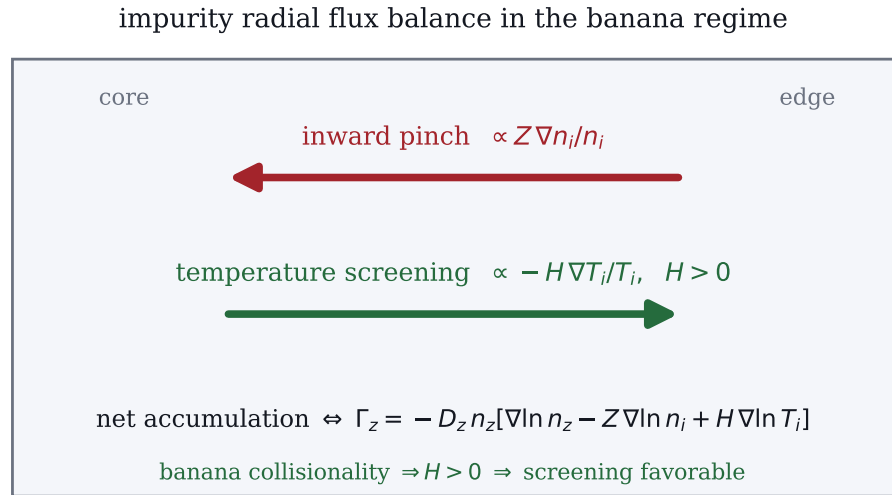


Figure 8. (Schematic.) Neoclassical impurity radial-flux balance in the banana regime: the inward pinch driven by the main-ion density gradient (red) competes against the outward temperature-screening flux (green). At the breeder core collisionality the screening coefficient $H > 0$, so a peaked ion-temperature profile opposes high- Z accumulation.

5.3 Coupling to the radiated-power budget

Temperature screening keeps impurities out; the collisional–radiative cooling curves say how much a given residual concentration would cost in radiated power. Independent ADAS collisional–radiative calculations (radas; serial BR-L2-A29 [13]) give the impurity cooling functions at the 15 keV core: $L_z^W = 5.7 \times 10^{-32} \text{ W m}^3$ for tungsten (mean charge $\langle Z \rangle = 61.9$) and $L_z^{\text{Ar}} = 1.45 \times 10^{-33} \text{ W m}^3$ for argon — tungsten radiating roughly forty times harder per ion in the core. Folding these against the frozen profiles bounds the tolerable concentration (figure 9): the tungsten sensitivity is $\approx 0.19 \text{ MW per ppm}$, so holding the core tungsten concentration to $c_W \lesssim 3 \text{ ppm}$ keeps the impurity radiated-power add-on below $\sim 0.6 \text{ MW}$ (a small fraction of the core radiated power), while argon — a far softer core radiator — is tolerable at the several-hundred-ppm level. The neoclassical screening regime and the radiative tolerance therefore tell a consistent story: the core physics favors impurity exclusion, and the radiative budget sets a

few-ppm ceiling on the high- Z residue the screening must hold. The core and edge radiated-power budget is closed in the companion collisional-radiative study [30].

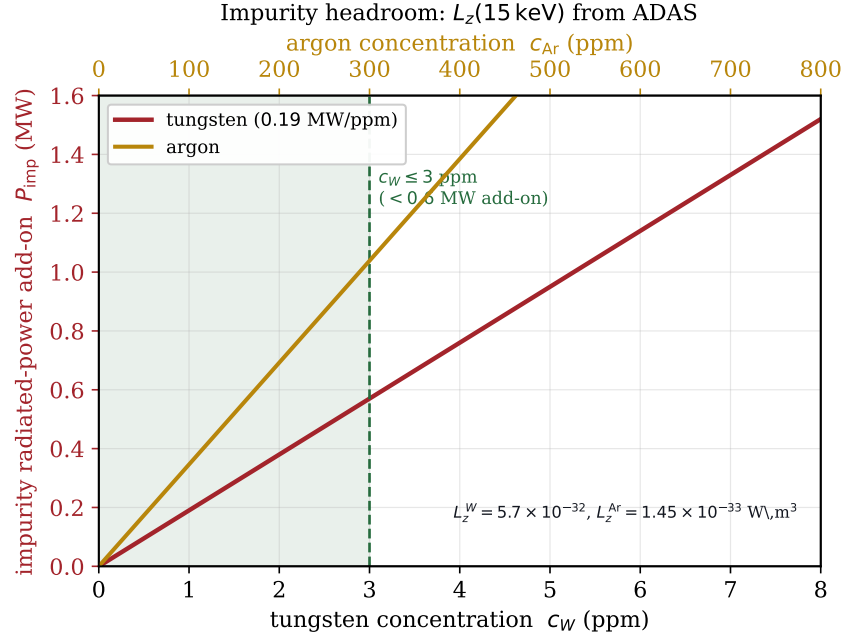


Figure 9. Impurity radiated-power headroom from ADAS collisional-radiative cooling curves (serial BR-L2-A29). Tungsten (red, lower axis) adds $\approx 0.19 \text{ MW ppm}^{-1}$; holding $c_W \lesssim 3$ ppm keeps the add-on below $\sim 0.6 \text{ MW}$. Argon (gold, upper axis) is a far softer core radiator and tolerable at hundreds of ppm.

6 GPU-accelerated HPC methodology

6.1 Codes run for this study

The neoclassical results were produced on the Kronos NVIDIA GPU HPC campaign, the same verification-grade computing programme that carries the breeder physics de-risking set. For *this* paper the primary solver is NEO (GACODE) [6, 7], run as a drift-kinetic local solve at three flux surfaces with the full linearized Fokker–Planck collision operator and kinetic electrons. The analytic-model comparison (Sauter [3], Koh [8], modified-Sauter and the Hirshman–Hinton banana limit [1]) is evaluated in the same GACODE theory module so that the drift-kinetic and analytic bootstrap currents share an identical geometry and normalization, and the impurity cooling functions are computed with radas over the ADAS ADF11 collisional-radiative data [13]. The turbulent transport this collisional baseline underlies was computed with the nonlinear gyrokinetic code CGYRO [16, 17] on the same campaign and is reported in the companion negative-triangularity study [25]; the neoclassical baseline and the turbulent transport are the two halves of the same confinement picture.

6.2 GPU acceleration, problem scale and verification

The neoclassical drift-kinetic solve assembles the block-sparse operator of equation (13) — with the velocity-space resolution $N_\varepsilon \times N_\xi \times N_\theta = 8 \times 17 \times 25$ per surface for a two-species (kinetic-electron, DT-ion) system — and factors it once with a sparse direct LU solver per surface; the neoclassical problem is modest in scale and memory-bound rather than throughput-bound, and it is the surrounding nonlinear-gyrokinetic and neutronics campaigns that consume the bulk of the GPU fleet’s throughput. The GACODE suite is GPU-accelerated across the campaign’s NVIDIA A100/H100/A10 nodes, and the neoclassical solves were carried within that environment so that the analytic-model comparison, the CGYRO turbulence runs and the neoclassical baseline all share a single, version-pinned toolchain. Reproducibility is enforced at the register level: the config-22021 anchor, the NEO input decks, and the driver that generates the three-surface sweep are archived with the deposit, the run is stamped with the exact GACODE build, and an independent re-run reproduces the NEO-to-Sauter ratio (register verdict: reproduced). The verification-first methodology — frozen anchors, independent-library gate reproduction, and a byte-for-byte provenance catalog — is described in the companion verification study [31]. No compute-cost

information is reported; the campaign is referred to only by the codes it ran and the scale of the problems it solved.

7 Discussion

7.1 The result against the neoclassical literature

The central quantitative finding — a drift-kinetic bootstrap current $0.81\text{--}0.87\times$ the Sauter analytic value in a deep-banana core — is exactly the direction and magnitude the neoclassical literature predicts. Comparisons of the Sauter formulas against dedicated numerical solvers have repeatedly shown the analytic fit to over-predict the bootstrap current at low collisionality and in strongly-shaped or steep-gradient regions, motivating the Redl *et al.* recalibration [5] and the Koh edge-pedestal treatment [8]. Our result reproduces that picture from first principles on a compact-ST operating point that the original Sauter database did not cover, and it quantifies the correction (a $\sim 17\%$ reduction at mid-radius) rather than merely flagging it. The trapped fractions $f_t = 0.685\text{--}0.727$ we compute are consistent with the large trapped populations expected at $A = 2.5$ and $\kappa = 2.0$, and they are what make the bootstrap current both large and sensitive to the trapped-passing physics that distinguishes NEO from the fits.

7.2 Impurity screening in context

The impurity picture is likewise consistent with the modern understanding of neoclassical impurity transport. The banana-regime temperature-screening mechanism [2, 10] predicts that a low-collisionality, temperature-peaked core opposes high- Z accumulation, and recent negative-triangularity experiments on DIII-D report flat or hollow core impurity profiles and impurity confinement times shorter than the energy confinement time [12] — i.e. favorable core impurity behavior in the same shape class as the Kronos point. Our computed core collisionality places the breeder in that screening-favorable regime, and the ADAS cooling-curve coupling turns the qualitative regime statement into a quantitative few-ppm tungsten ceiling. The consistency of the two independent lines — the drift-kinetic screening regime and the collisional-radiative radiated-power budget — is the point: they agree that the compact breeder core is a favorable, not adversarial, setting for high- Z control, with a well-defined tolerance.

7.3 Consequences for the design point

Grounding the bootstrap current in kinetic theory changes it from an assumption into a measured quantity with a stated uncertainty. The current-drive burden implied by $f_{bs} = 0.155$ — an external-drive requirement of $\simeq 8.2$ MA — now rests on a calculation whose model uncertainty is bounded by the solve rather than by the factor-of-two spread across analytic fits, and the companion current-drive and solenoid-free-startup studies [27, 28] inherit that bounded baseline. The same framework carries the poloidal flows and the temperature-screening mechanism, so the impurity-transport question sits on the same first-principles footing. The collisional baseline is now in place beneath the turbulent transport that sets the confinement.

8 Limitations

One honest gate bounds the claims. The bootstrap *ratio* to Sauter is a clean first-principles result: it is a local, geometry-consistent comparison at each surface, insensitive to the profile-integration and normalization approximations. The integrated *magnitude* $f_{bs} = 0.155$ is a different object. A crude three-surface profile integral of the local NEO current, with the flat density that suppresses the ∇n drive and an approximate field normalization, returns $f_{bs} \sim 0.075\text{--}0.087$ — roughly a factor of two below the toolkit anchor. This is not a refutation of 0.155: the discrepancy is within what the flat-density profile ($\partial \ln n / \partial r = 0$ kills a real gradient drive), the three-point integration, the $\langle j_{\parallel} B \rangle / B_{\text{unit-as-}j_{\parallel}}$ approximation, and the approximate B_{unit} can explain, and the integrated toolkit value carries the geometry the local mapping drops. The confirming calculation for the absolute fraction — and for the quantitative impurity flux Γ_z — is a full-profile, multi-species NEO run from a converged free-boundary equilibrium (an `input.gacode` built from the FreeGS/CHEASE geometry for $\delta = -0.30$ [19, 18], with proper flux-surface integration of the toroidal current). That run is the natural follow-on; the present study fixes the ratio, the regime, and the sign of the screening, which are the load-bearing conclusions.

9 Conclusion

A drift-kinetic NEO solve on the negative-triangularity Kronos breeder point returns a parallel bootstrap current $0.81\text{--}0.87\times$ the Sauter analytic value across the mid-radius (0.843 at $r/a = 0.6$),

reproducing the de-risking register’s $\sim 83\%$ -of-Sauter anchor from first principles and explaining the offset as the expected banana-regime correction at $\nu_*^e \approx 0.012$ — a correction that shrinks monotonically toward unity as collisionality rises outward, the signature of real low-collisionality physics rather than a numerical artifact. The same solve delivers the large trapped fractions of the compact, strongly-shaped ST ($f_t = 0.685\text{--}0.727$), the neoclassical poloidal flows, and the temperature-screening mechanism, and it places the breeder core in the screening-favorable banana regime; coupling that regime to independent ADAS collisional–radiative cooling curves bounds the tolerable tungsten concentration at a few ppm. With the bootstrap ratio anchored in kinetic theory, the integrated fraction pinned at $f_{bs} = 0.155$ pending a full-profile confirmation, and the impurity screening quantified, the neoclassical transport of the compact breeder is now a grounded collisional baseline — with a stated model uncertainty — beneath the turbulent transport that sets the confinement.

Acknowledgments

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Reproducibility. The neoclassical results are frozen anchors in the Kronos Physics De-Risking Register [24] (doi:10.5281/zenodo.22133057): the first-principles NEO drift-kinetic solve (BR-L2-A16), the integrated bootstrap fraction (BR-L2-A6), and the ADAS impurity cooling curves (BR-L2-A29). They were computed with NEO (GACODE) and radas on the config-22021 anchor and are regenerable from the archived driver, NEO input decks and version stamp deposited with this paper; this paper is archived at doi:10.5281/zenodo.22645832; official version: <https://www.kronosfusionenergy.com/publications/neoclassical-transport-poloidal-flows-and-impurity-scre.html>.

Data availability

The NEO inputs and outputs, the analytic-model comparison, and the ADAS cooling-curve data used for the figures are archived with the deposit at doi:10.5281/zenodo.22645832, which references the Kronos 2026 series. All quantitative values trace to the register [24] by the serial identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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